

## Section A: Multiple Choice Questions (30 points)

All questions in this section are worth 2 points each.

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|-----------------|-----------------|-----------------|------------------|------------------|
| 1. <u>  A  </u> | 4. <u>  B  </u> | 7. <u>  B  </u> | 10. <u>  C  </u> | 13. <u>  B  </u> |
| 2. <u>  A  </u> | 5. <u>  A  </u> | 8. <u>  B  </u> | 11. <u>  A  </u> | 14. <u>  D  </u> |
| 3. <u>  A  </u> | 6. <u>  D  </u> | 9. <u>  C  </u> | 12. <u>  B  </u> | 15. <u>  C  </u> |

## Section B: Short Answer Questions (32 points)

All questions in this section are worth 4 points. There is no partial credit.

- 9 teeth
- 5.4
- 1.11 MJ
- 102.6 seconds
- 1.936
- 0.875
- 14.08
- $5\pi$  or 15.71

## Section C: Free Response Questions (38 points)

All point values are provided in the exam. If other methods are correctly used, give points for writing the equation and solving for the answer separately. Partial credit can be given for a FBD. If an incorrect value is calculated for a, partial credit will not be awarded for the “correct” but incorrect answer for b. Sorry :(

- (19 points)
  - Either uses conservation of energy ( $mgh = \frac{1}{2}mv^2$ ) or traditional kinematics ( $v_f^2 = v_i^2 + 2a\Delta x$ ) [1]. Substitutes correctly and obtains either  $v = \sqrt{2gh} = 6.26$  m/s or  $v_f = \sqrt{2g \sin \theta \cdot \frac{2}{\sin \theta}} = 6.26$  m/s [2].
  - Either uses conservation of energy and sets work done by friction to change in kinetic energy ( $F\Delta x = \frac{1}{2}m(\Delta v)^2$ ) or traditional kinematics ( $v_f^2 = v_i^2 + 2a\Delta x$ ) [1]. Substitutes correctly and obtains either  $x = \frac{v_i^2}{2a}$  or  $x = \frac{mv_i^2}{2F}$  [1]. Obtains final answer of 6.66 meters [2]. Note that  $F = mg \sin \theta$  and  $a = \frac{F}{m}$ .
  - Identifies that the force along the ramp downwards is  $F = mg \sin \theta$  [1] and that the force of friction is  $\mu_s \cdot N$  or  $\mu_s \cdot mg \cos \theta$  [1]. Sets the two equal [1] and finds that  $\mu_s = \frac{mg \sin \theta}{mg \cos \theta}$  [1]. Obtains final  $\mu_s = 0.445$  [2]. Alternatively, states that  $\mu_s = \tan \theta = 0.445$  [6].
  - Uses  $v_f^2 = v_{base}^2 + 2a\Delta x$  [1] to find  $v_{base} = \sqrt{2 \cdot 5 \cdot a} = 5.42$  m/s [1]. Identifies that acceleration of the block down the ramp is  $mg \sin \theta - \mu_k mg \cos \theta$  [2] and solves for  $\mu_k = 0.111$  using  $v_{base}^2 = v_i^2 + 2a\Delta x$  [2].

## 2. (19 points)

- (a) Sets up free body diagrams for both masses, getting  $F_{net1} = T - 3g$  and  $F_{net2} = T - 8g$  [1]. Notes that  $a_1 = -a_2$ , and thus sets  $\frac{F_{net1}}{m_1} = -\frac{F_{net2}}{m_2}$  [1]. Solves for  $T = 42.81N$  [1]. Finally, solves for  $a_1 = 4.46m/s^2$  and  $a_2 = -4.46m/s^2$  [1]. If signs of acceleration are not opposite, [0.5].
- (b) Uses  $\Delta x = v_i t + \frac{1}{2}at^2 = 0.5gt^2$  [1]. Solves for  $t = \sqrt{\frac{1}{g}}$  [1]. Obtains  $t = 0.319$  seconds [1].
- (c) Calculates the “available energy of the system” by calculating change in gravitational potential energy using  $\Delta PE = -m_1gh + m_2gh$  [1]. Finds  $\Delta PE = 24.5$  Joules [1]. Sets this equal to the change in gravitational potential energy  $m_3gh$  [1]. Solves correctly, obtaining  $h_{bowling} = 0.417$  meters and  $h_{bouncy} = 12.5$  meters [2].
- (d) Mass 1:  $T$  upwards,  $Mg$  or  $Mg_k$  downwards [1]. Mass 2:  $2T$  upwards,  $2Mg$  or  $2Mg_k$  downwards [1]. If they create a  $T_2$  upwards, [0.5]. Mass 3:  $T$  upwards,  $3Mg$  or  $3Mg_k$  downwards [1].
- (e) Uses free body diagrams to set up three equations,  $T - Mg = Ma_1$ ,  $2T - 2Mg = 2Ma_2$ , and  $T - 3Mg = 3Ma_3$  [1]. Uses the “conservation of string” to get the equation  $a_1 + 2a_2 + a_3 = 0$  [1]. Rearranges the first three equations into  $a_1 = \frac{T}{M} - g$ ,  $a_2 = \frac{T}{M} - g$ , and  $a_3 = \frac{T}{3M} - g$  [1]. Combines with “conservation of string” to get  $(\frac{T}{M} - g) + 2(\frac{T}{M} - g) + \frac{T}{3M} - g = 0$ . Then,  $\frac{10T}{3M} - 4g$ , and finally obtains  $g_k = \frac{5T}{6M}$  [1].
- (f) Substitutes  $T = 36$  N and  $M = 5$  kg, obtaining  $g_k = \frac{5*36}{6*5} = 6$  m/s<sup>2</sup> [1].